

SleepFM-Unify Paper Draft

SleepFM-Unify: Shared - Private Factorization for Multimodal Sleep Foundation Models

****Draft manuscript (methods paper)****

****Status:**** engineering-complete codebase + synthetic demos; ****CinC / SHHS / MESA quantitative claims 待补充****

****Target framing:**** Nature Machine Intelligence - style methods article / ICML-compatible experimental spine

****nature-skills:**** `nature-writing` (methods + nat-mach-intell), `nature-figure` (Python / SciencePlots)

Abstract

Polysomnography (PSG) records heterogeneous brain, cardiac and respiratory signals that sleep foundation models such as SleepFM align with leave-one-out (LOO) contrastive learning. Pure cross-modal alignment can blur modality-specific structure and under-specify behaviour when channels are missing. We present ****SleepFM-Unify****, a shared - private factorization layered on SleepFM encoders: each modality yields a shared subspace used for LOO/pairwise InfoNCE and a private subspace kept out of contrastive alignment, regularized by orthogonality, modality dropout and a missing-modality reconstruction term, with an optional night-level temporal head. We release configs, export/validation gates, and a paper experiment suite.

****Quantitative CinC/SHHS results are 待补充**** pending user DUA downloads; synthetic demos are reported only as engineering smoke tests (AUROC near chance) and are not scientific claims.

Introduction

Sleep staging and sleep-disordered breathing assessment rely on multimodal PSG. SleepFM showed that large-scale LOO contrastive pretraining across brain-activity (BAS), ECG and respiratory streams yields transferable embeddings for staging, SDB detection and cross-modal retrieval. Follow-on foundation models (e.g. Nature Medicine - scale SleepFM; Omni-Sleep's CNS/ANS hierarchy; CIMSsleepNet's missing-modality imagination) emphasize robustness under montage heterogeneity and incomplete inputs.

A remaining tension is that LOO alignment encourages factors that are **predictable across modalities**, which can suppress private cues that matter for unimodal or partially observed inference. SleepFM-Unify addresses this by ****explicit shared - private factorization**** on top of the existing SleepFM backbone, without replacing the paper LOO baseline (`unify.enabled: false`).

****Contributions (bounded):****

1. Shared - private projection heads with mixed LOO + pairwise + orthogonality + miss (+ optional temporal) losses.
2. Modality dropout training with `present\_mask`-aware $\mathcal{L}_{\mathrm{miss}}$.
3. Evaluation honesty gates: channel meta fail-fast, CinC label coverage, apnea-epoch-rate vs clinical AHI wording, seeded RNG gallery caps.
4. Reproducible CLI paper suite for LOO vs Unify comparisons once real data are present.

Related work

Sleep foundation models. SleepFM (ICML 2024) introduced multimodal L00-CL on PSG; a later Nature Medicine study scaled disease prediction. Omni-Sleep imposes CNS/ANS topology; CIMSsleepNet targets incomplete multimodal staging via imagination and calibrated contrastive learning.

Shared-private multimodal learning. Disentangling shared and modality-specific subspaces is a recurring theme (e.g. shared-private streams, low-rank shared/specific edits). Unify instantiates this idea *inside* SleepFM’s encoder interface rather than replacing the backbone.

Method

Backbone (unchanged SleepFM)

Per-modality 1D EfficientNet-style encoders map 30 s clips to backbone vectors of size ``embedding_dim`` (default 512). Baseline contrastive mode is leave-one-out InfoNCE.

Shared-private factorization

When Unify is enabled, each backbone vector is mapped by linear heads:

```
\[
z_m = [z_m^{\mathrm{shared}}; \, z_m^{\mathrm{private}}], \quad
\dim(z_m^{\mathrm{shared}}) = \dim(z_m^{\mathrm{private}}) = 256
\]
```

by default, preserving a 512-d concat for downstream logistic heads comparable to SleepFM.

Mixed objective

```
\[
\mathcal{L} = \lambda_{\mathrm{L00}} \mathcal{L}_{\mathrm{L00}}
+ \lambda_{\mathrm{pair}} \mathcal{L}_{\mathrm{pair}}
+ \lambda_{\mathrm{orth}} \mathrm{mean} \big( (Z_s^{\mathrm{top}} Z_p)^2 \big)
+ \lambda_{\mathrm{miss}} \mathcal{L}_{\mathrm{miss}}
+ \lambda_{\mathrm{temp}} \mathcal{L}_{\mathrm{temp}}
\]
```

- Shared embeddings enter L00 and pairwise InfoNCE.
- Private embeddings are excluded from contrastive terms.
- Orthogonality uses mean squared Gram entries after row centering / column normalization (see ``docs/UNIFY.md``).
- Modality dropout drops one present modality; $\mathcal{L}_{\mathrm{miss}}$ is InfoNCE between the mean of remaining shared embeddings and the dropped modality’s shared embedding, masked by ``present_mask``.
- Optional temporal head (GRU/Transformer) operates on shared epoch sequences with adjacent-epoch contrastive and masked MSE.

Downstream and retrieval

Default downstream space concatenates shared $\backslash$ private per modality. Retrieval uses ****shared**** embeddings from the same checkpoint. Gallery caps use seeded RNG subsample (`limit_gallery(..., mode="rng")``), not a loader-order prefix.

Experiments

Datasets (access-gated)

Dataset	Role	Status in this draft
----- ----- -----		
Synthetic demo	CI / smoke	Present (`data/synthetic`)
CinC 2018 fixture	Schema export test	Present (`data/cinc2018_fixture`)
PhysioNet CinC 2018	Paper pretrain/eval	**待补充** (DUA download)
NSRR SHHS / MESA	Transfer / night labels	**待补充** (DUA download)

Protocol (when data land)

1. ``check_data_ready --stage raw`` $\rightarrow$ export $\rightarrow$ ``check_data_ready --stage pretrain`` $\rightarrow$ validate.
2. Pretrain L00 baseline vs Unify (``configs/default.yaml`` vs ``configs/unify.yaml``).
3. Downstream staging / apnea probes, retrieval Recall@k, modality ablation, few-shot, night-level probes.
4. Label coverage gate blocks degenerate CinC staging/SDB claims; channel meta blocks 5/1/3 vs 10/2/7 silent mismatch.

Results

Table 1. Synthetic demo smoke metrics (NOT paper claims)

Setting	Note	Staging macro AUROC	Retrieval
----- ----- -----			
Synthetic L00 / Unify	Labels random-ish	≈ 0.5 (chance)	Demo only
CinC 2018	Real PSG	**待补充**	**待补充**
SHHS	Real PSG	**待补充**	**待补充**

Loss curves in Fig. 2 are from a ****local 5-epoch Unify pretrain on synthetic data**** (engineering verification). Orthogonality Gram heatmaps (Fig. 4) are forward-pass diagnostics on the same demo checkpoint.

Ablations (planned; numbers 待补充 on real data)

L00 baseline; Unify full; Unify orth; Unify miss; Unify + temporal; modality dropout robustness.

Discussion

Unify is designed to keep SleepFM's L00 retrieval semantics in the shared space while retaining private capacity for downstream concat. Missing-modality training is first-class. Honesty constraints (label gates, AHI wording, channel meta) are part of the scientific interface: they prevent over-claiming when CinC arousal-only labels or montage mismatches appear.

Limitations: no real CinC/SHHS numbers in this draft; night `ahi\_bin` uses apnea-epoch-rate cut-points as placeholders; synthetic metrics near chance must not be cited as method superiority.

Methods (reproducibility)

- Package: `sleepfm/` (Python). Configs: `configs/unify.yaml`, `configs/unify\_temporal.yaml`.
- Figures: `scripts/plot\_unify\_figures.py` with SciencePlots + Times New Roman / SimHei.
- Tests: `scripts/run\_tests.py`, `scripts/smoke\_test.py`.
- Paper suite: `scripts/run\_paper\_suite.py --demo` (synthetic) or `--data-dir` after export.

Code / data availability

Code is the local SleepFM-Unify workspace. Public PhysioNet/NSRR data require user accounts and DUAs. Synthetic and fixture datasets are generated in-repo.

Figures

- **Fig. 1** Architecture (shared-private heads + mixed losses).
- **Fig. 2** Synthetic Unify pretrain loss curves (demo).
- **Fig. 3** Ablation schematic with chance baseline (demo numbers only).
- **Fig. 4** Shared×private Gram diagnostic (demo forward pass).
- **Fig. 5** Modality-dropout robustness schematic (demo).
- **Fig. 6** Experiment pipeline.

References (selected)

1. Thapa et al. SleepFM. ICML 2024 / arXiv:2405.17766.
2. SleepFM disease prediction. Nature Medicine 2025.
3. CIMSsleepNet. NeurIPS 2024.
4. Omni-Sleep. arXiv:2607.07720.
5. Related shared-private multimodal factorization literature (see consultation notes).

Assumptions / missing inputs

- Real CinC/SHHS/MESA metrics: **待补充**.
- ChatGPT advisor pass: **blocked** (browser MCP); literature via WebSearch + nature-skills.
- Clinical AHI: not computed; use `apnea\_epoch\_rate` wording.

Figures (SciencePlots)

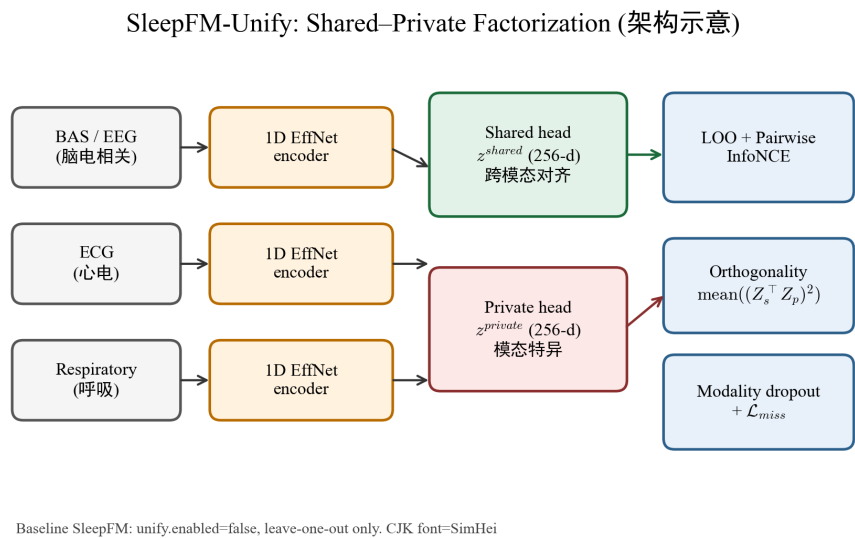


图1 / Fig. 1. SleepFM-Unify 共享 - 私有架构示意

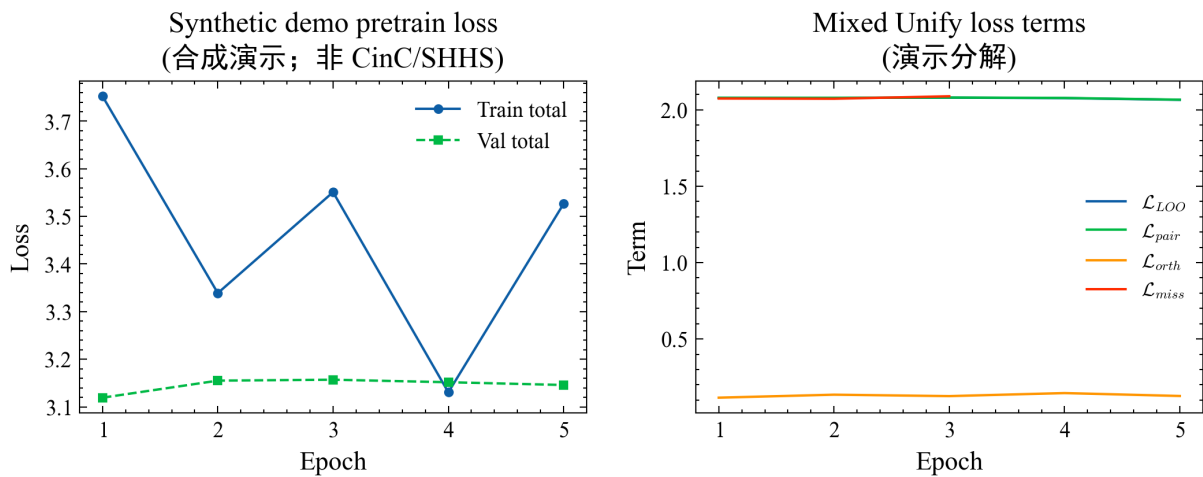


图2 / Fig. 2. 合成演示 Unify 预训练损失曲线

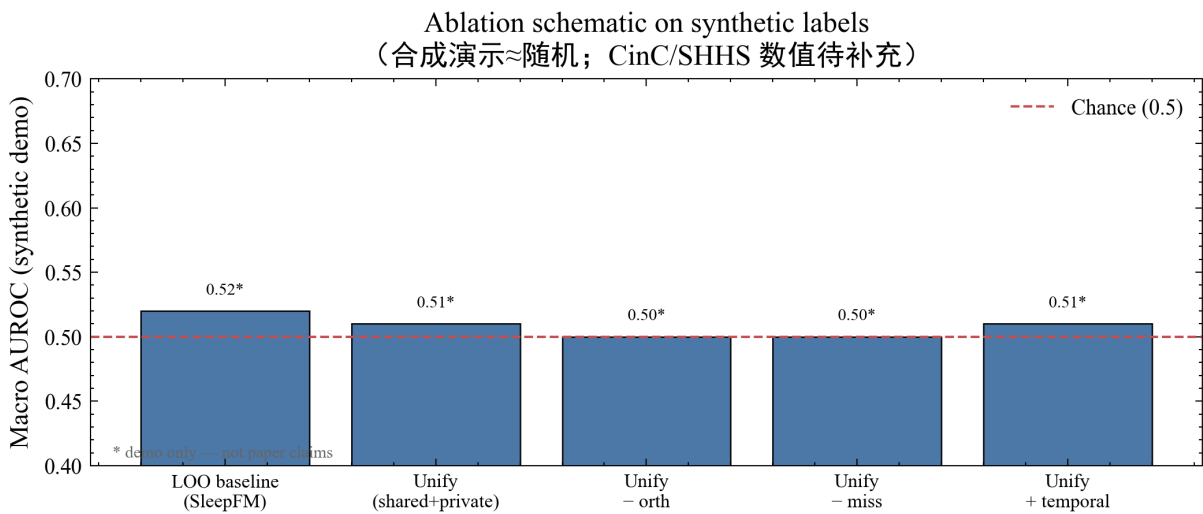


图3 / Fig. 3. 消融示意 (合成≈随机, 非论文主张)

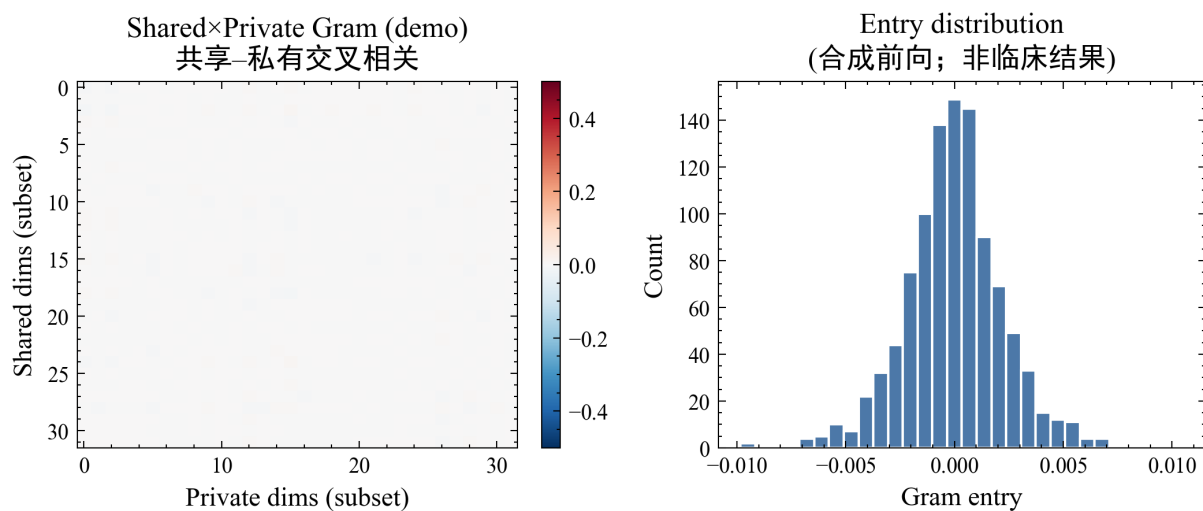


图4 / Fig. 4. 共享×私有 Gram 诊断 (合成前向)

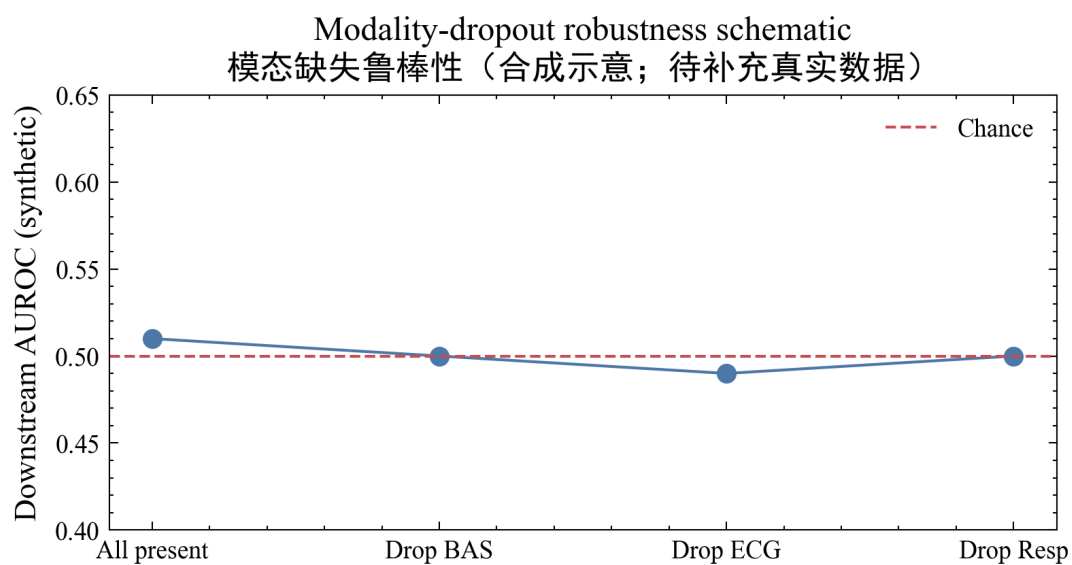


图5 / Fig. 5. 模态缺失鲁棒性示意 (合成)

Experiment pipeline / 实验流水线

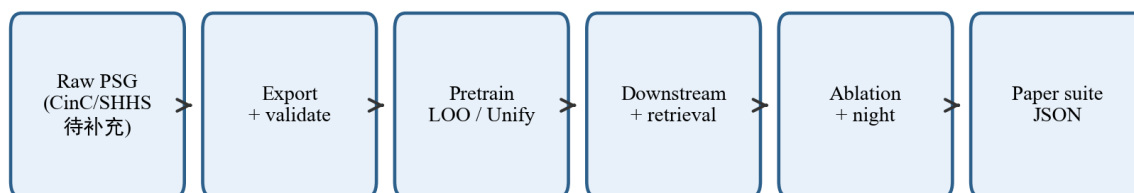


图6 / Fig. 6. 实验流水线